



Centre for Semiconductor Technologies (SEMIX)

3rd Annual SemiX Summit

Sep 25 - 26, 2025



Post Summit AIPC Workshop: Use Case

Presenter:

Udayan Ganguly: Professor In Charge SemiX

Nilesh Dange: Intel

Sumedh Risbud: Intel

Nitin Sarangdhar: Executive SemiX

September 27, 2025



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Nitin Sarangdhar: Executive, SemiX

Retired Founder, CEO NitAI

Retired Senior PE, Intel

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Today's Agenda



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- Use Case: Helmet Detection/License plate Image detection
- Overview of Image Detection
 - Perceptron
 - LeNet, MNIST dataset
 - Resnet, ImageNet-1K dataset
 - Yolov5, COCO dataset
- AI/PC based **Hands on activity**
 - Resnet: Golden Retriever Classification
 - Yolov5: Helmet Detection
- Lunch Break (around 1pm)
- Analysis of AI/PC results and mapping to CPU, GPU, NPU HW
- Google Colab Server based **Hands-on activity**
 - Yolov5 model training
- Hackathon use case discussion, timeline etc

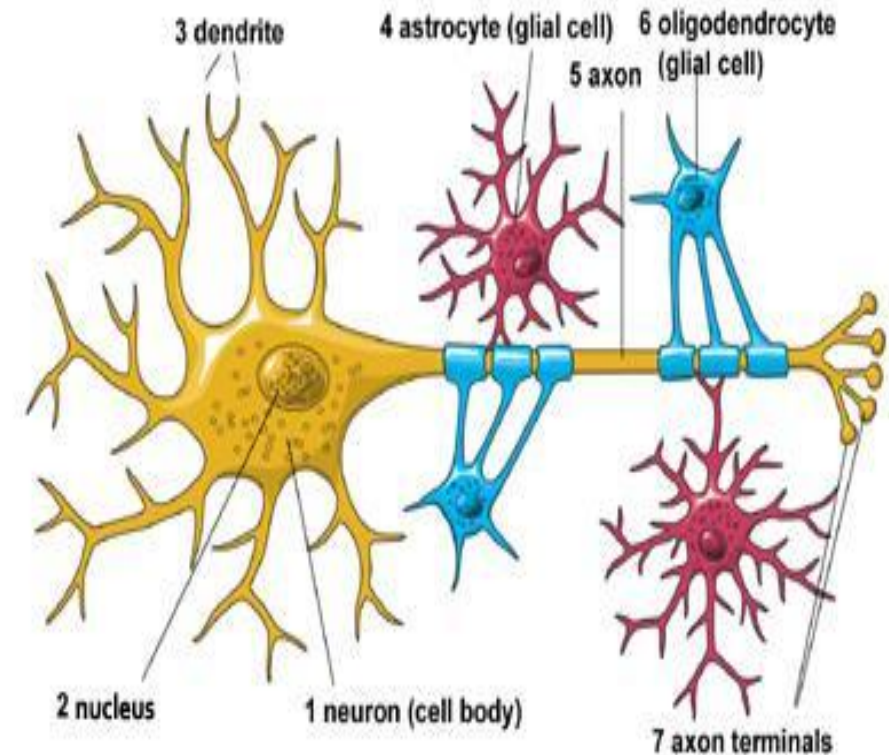


Biological Neuron



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- **Neurons:** Nerve cells that send messages all over your body to allow you to do everything
- **Synapse:** Communication channel between Neurons
- **Synaptic Plasticity:** Synaptic plasticity refers to the ability of synapses to strengthen or weaken over time, in response to increases or decreases in their activity. This is the fundamental mechanism of learning and memory in humans.





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Inspiration from the brain

- Human brain = billions of neurons firing, inspiring computer scientists.
- The question: *Can machines learn like humans?*
- Birth of **Perceptron**
- Transition to neural networks

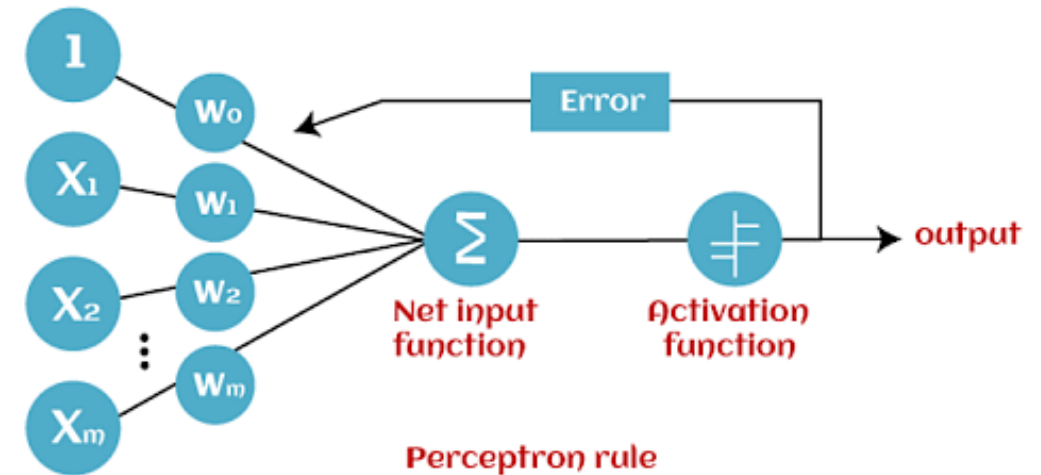


Perceptron



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- **Perceptron**
 - A basic building block of machine learning, inspired by a biological neuron.
- **Inputs and Weights**
 - Each input has a weight, similar to a biological synapse that controls signal strength.
 - The perceptron computes a weighted sum of its inputs.
- **Activation**
 - The weighted sum is passed through an activation function (e.g., step function) to produce an output.
- **Learning**
 - Learning occurs by adjusting the weights during training, analogous to synaptic plasticity in biological neurons.



The Perceptron (1950s-60s)

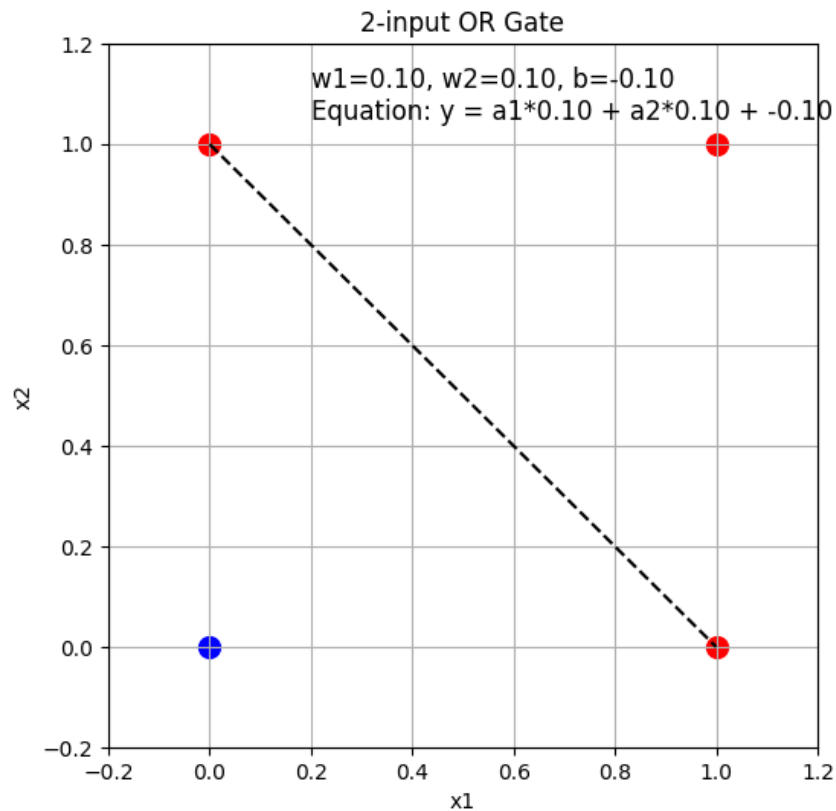


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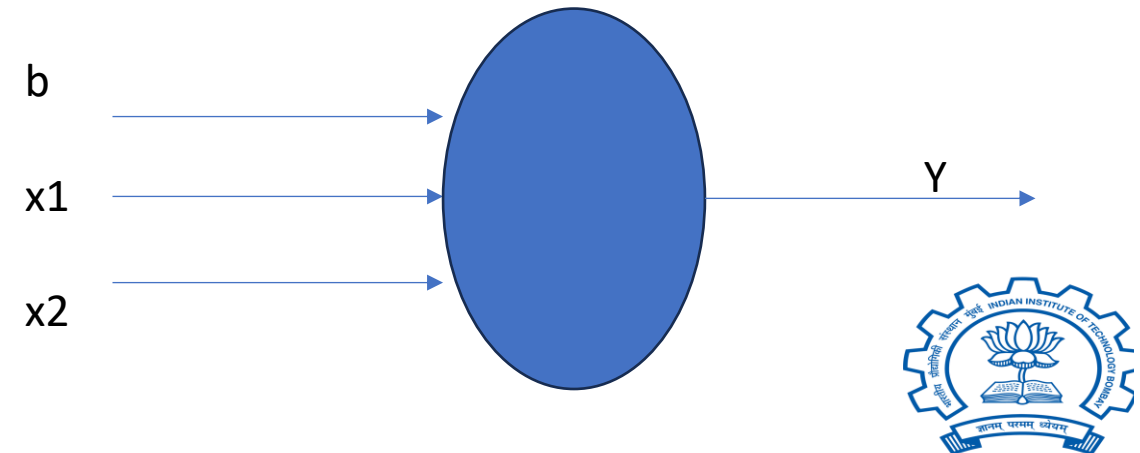
- Simple model of a “neuron”.
- This is one of the linear classifier algorithms
- It can classify simple patterns such as AND gate and OR gate
- Able to recognize **basic patterns**.



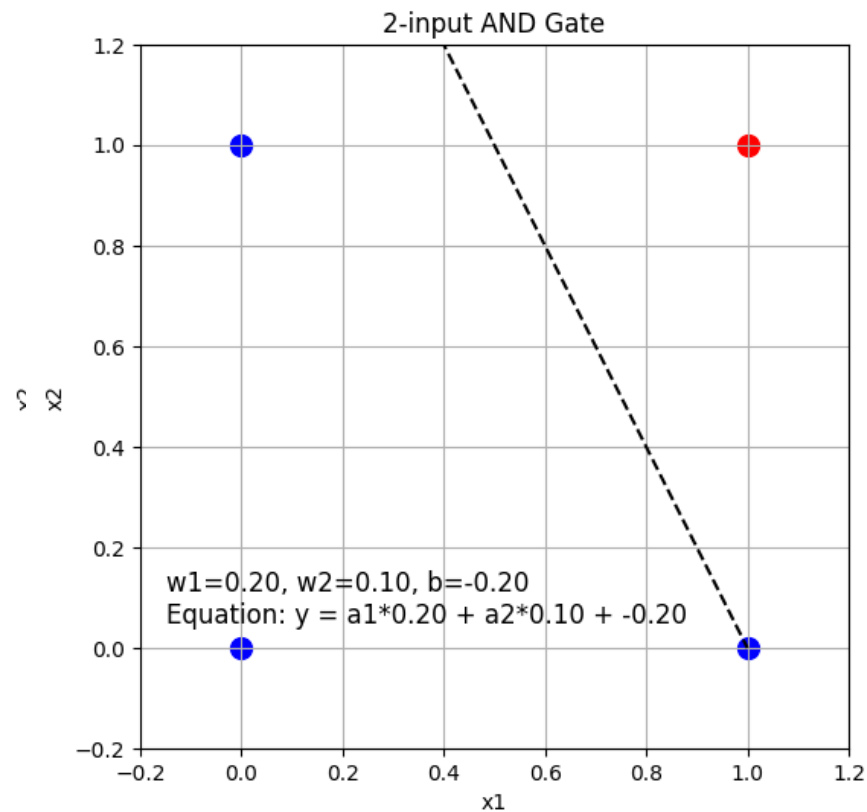
OR2 classifier using perceptron



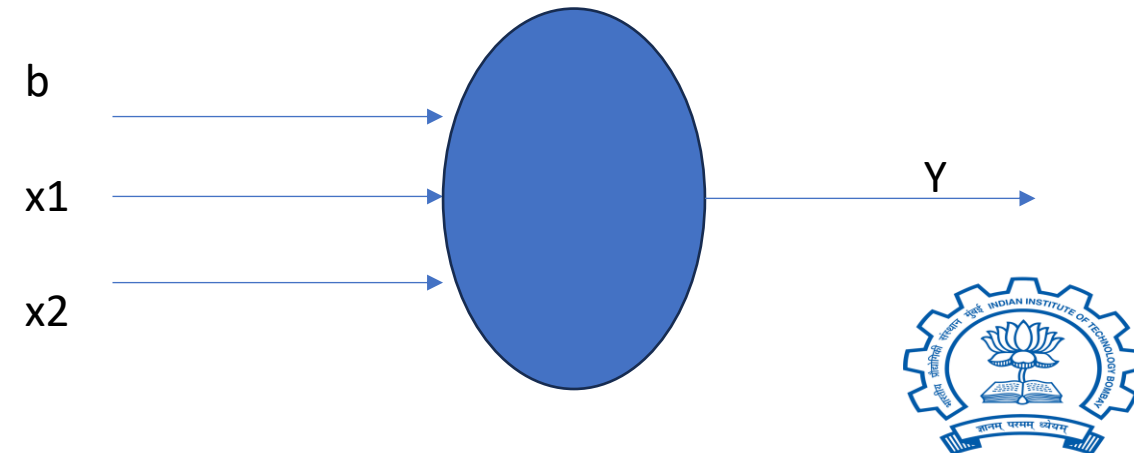
- 2 Input OR Gate
- $y = x1 \text{ OR } x2$
- $y = 1$ when $x1$ is 1 or $x2$ is 1
- $Y = 0$ when $x1$ is 0 and $x2$ is 0



AND2 classifier using perceptron



- 2 Input AND Gate
- $y = x1 \text{ AND } x2$
- $y = 1$ when both $x1$ and $x2$ are 1
- $Y = 0$ when $x1$ is 0 or $x2$ is 0



Perceptron Limitations



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- Banks needed to read checks → AI for digit recognition.(1980s)
- Original NIST database of handwritten digits was approx. 128x128 pixels per image
- The Perceptron based classifier provided about 50% accuracy on the NIST database. (Did not work well)



Simple Image Classification (1990s)



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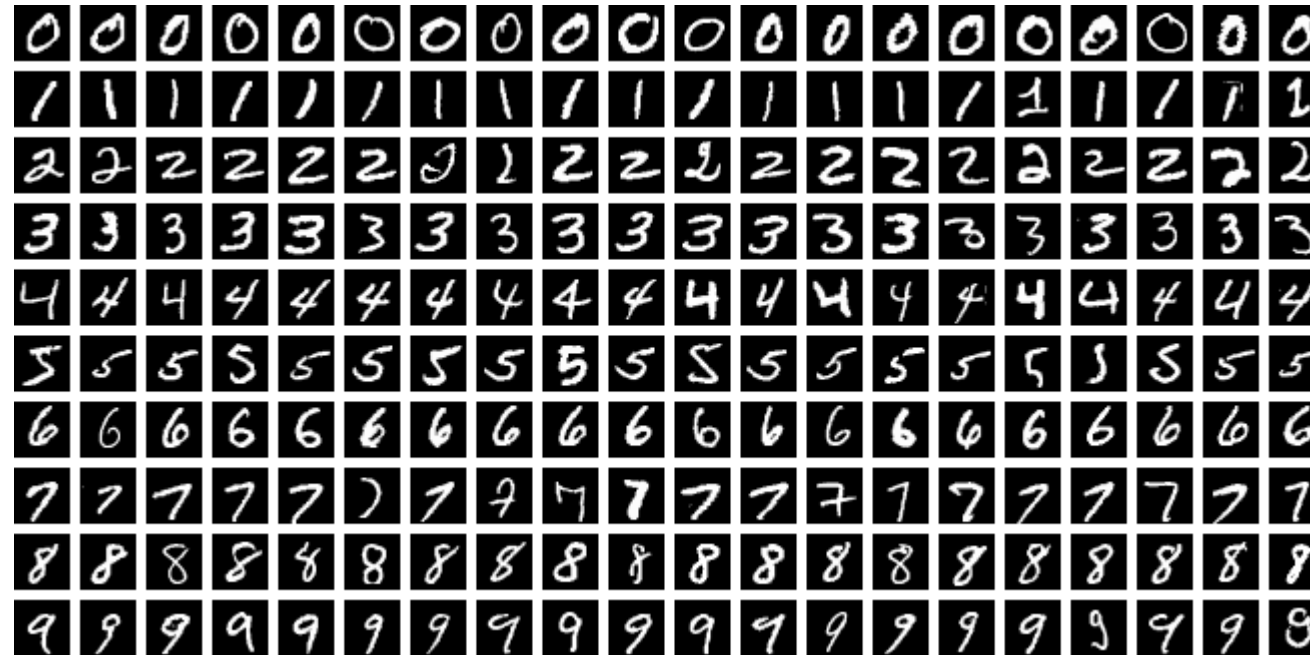
- **Convolutional neural networks (CNNs)** by Yann LeCun (**LeNet**). First real-world success of deep learning. (Some inspiration from Visual Cortex)
- MNIST (Modified NIST) database was created by Yann Lecun et al.
- This was based on image size of 28x28 pixels and a 8-bit gray scale value (0-255) The dataset has approx. 70000 image samples



MNIST Dataset



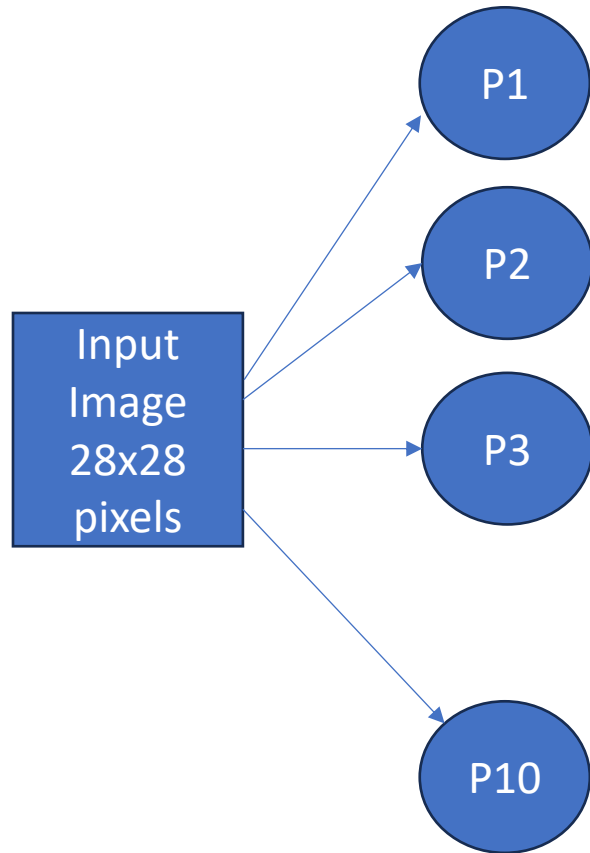
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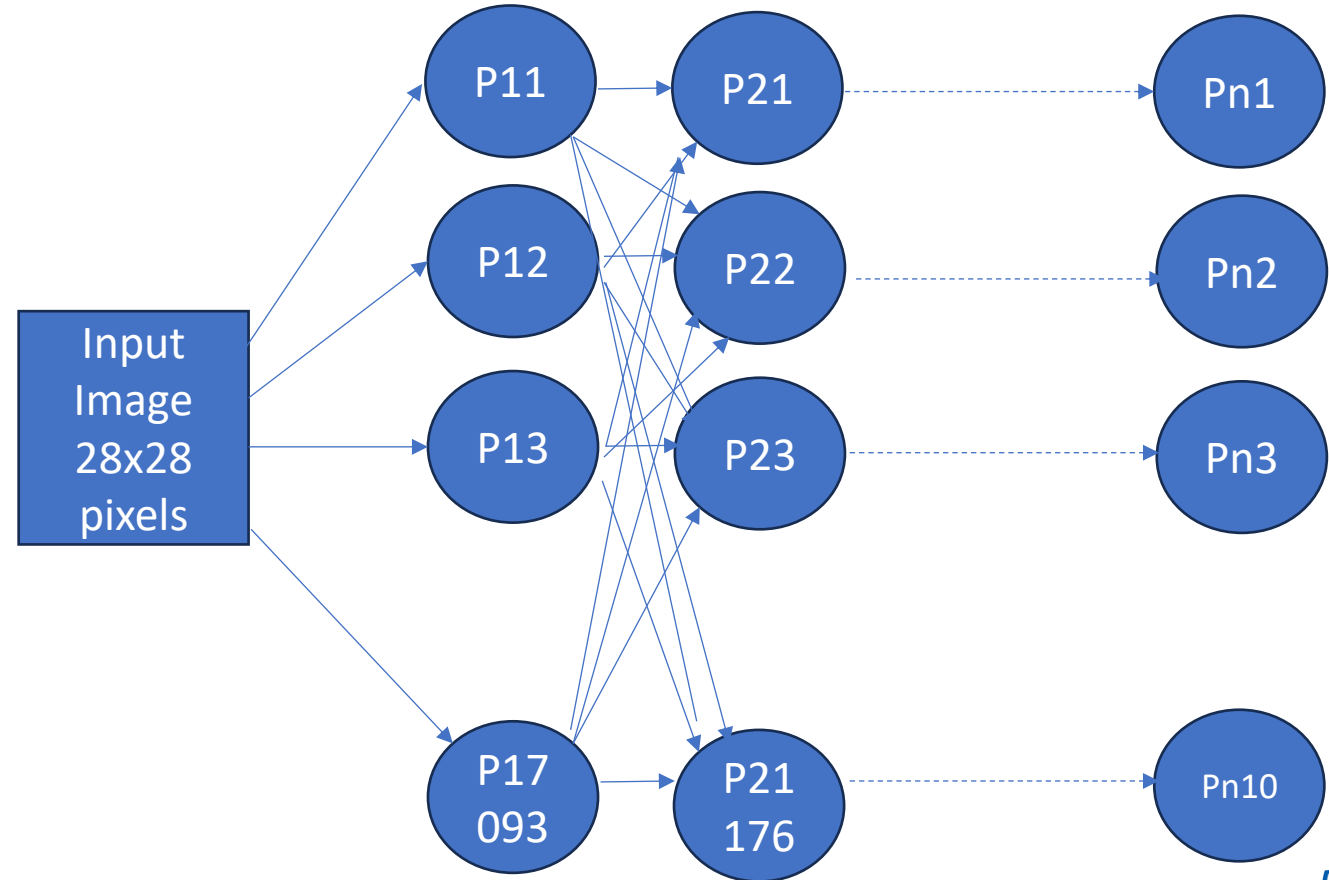
LeNet (Multi-Layer Perceptron)



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Perceptron



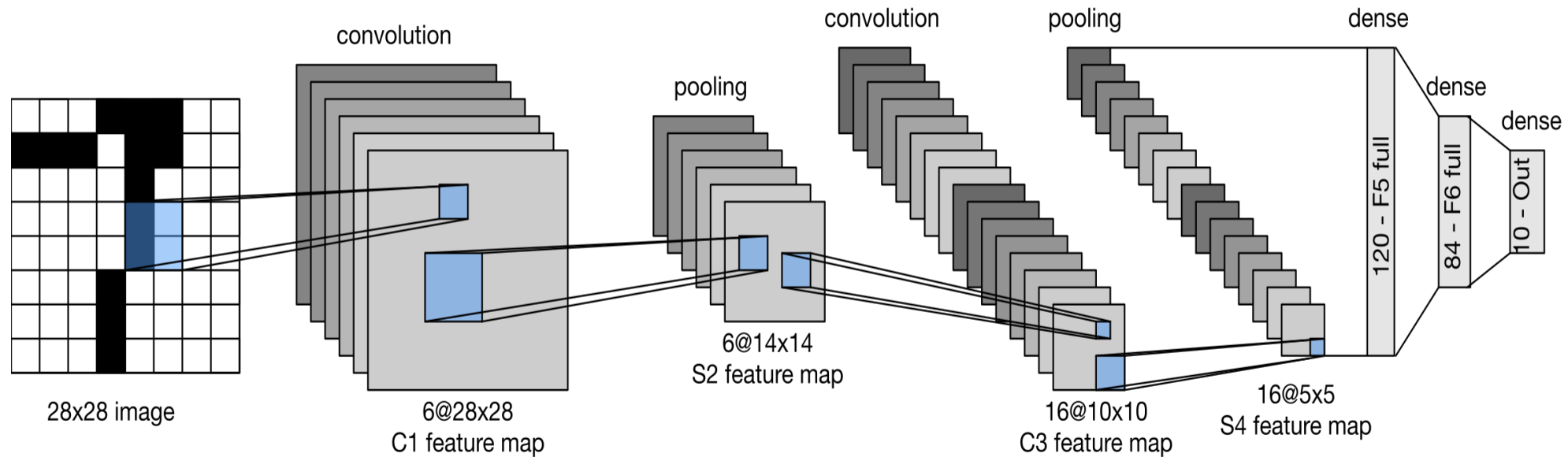
LeNet (Multi Layer Perceptron)



LeNet DataFlow



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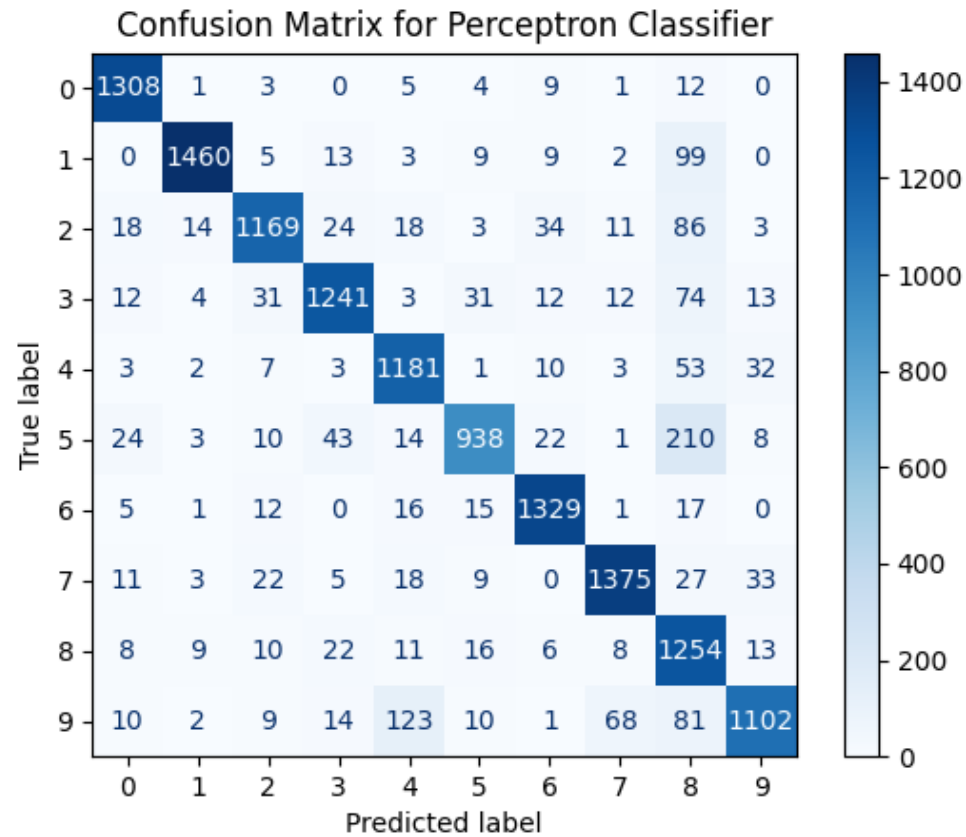
Data flow in LeNet. The input is a handwritten digit, the output is a probability over 10 possible outcomes. LeNet has **5–7 layers** and **~60,000 parameters**.



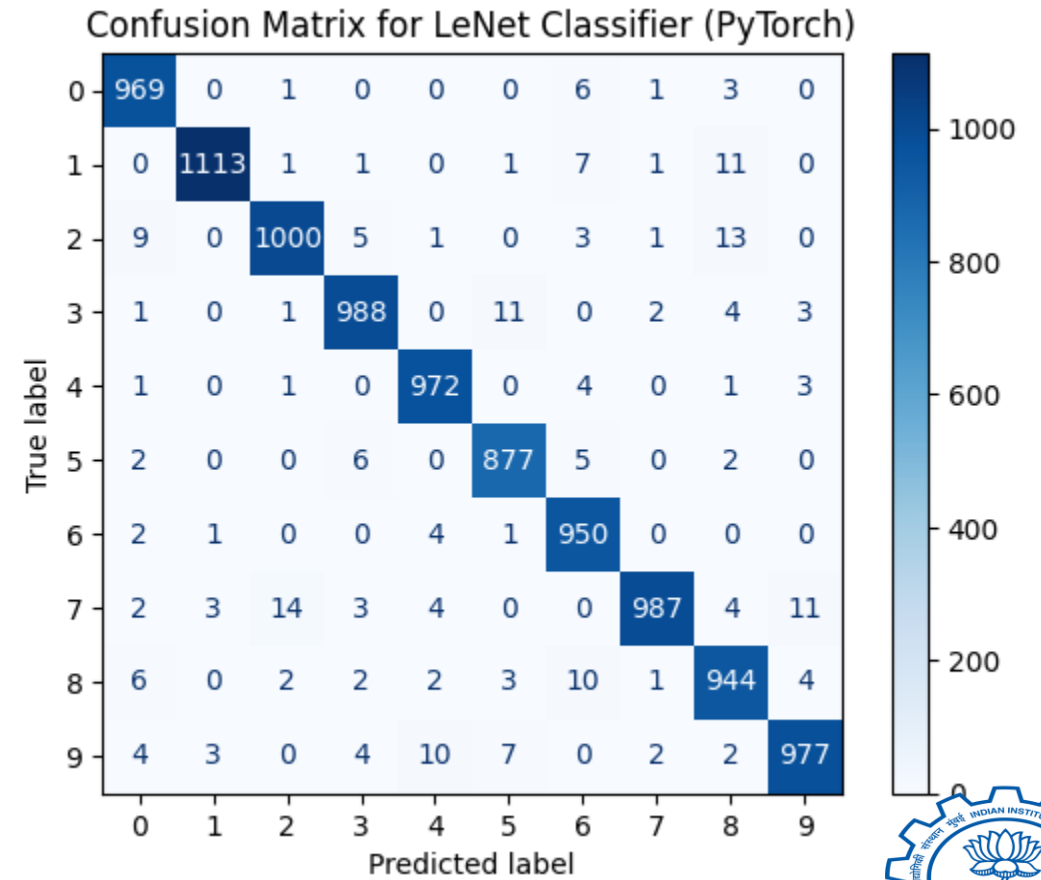
LeNet Details

- **Input:** 32×32 grayscale image
- **C1 – Conv:** 6 filters $5 \times 5 \rightarrow 28 \times 28$ feature maps
- **S2 – Pool:** 2×2 average pooling $\rightarrow 14 \times 14$
- **C3 – Conv:** 16 filters $5 \times 5 \rightarrow 10 \times 10$ feature maps
- **S4 – Pool:** 2×2 average pooling $\rightarrow 5 \times 5$
- **C5 – FC Conv:** 120 neurons (5×5 conv $\rightarrow$ fully connected)
- **F6 – FC:** 84 neurons
- **Output:** FC: 10 neurons (digit classes, softmax)

Perceptron/LeNet Accuracy Comparison



Classification accuracy: 88%



Classification accuracy: 97.77%

Modern Image Classification

- **ImageNet-1K**
- Released in 2009 as part of the **ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**.
- **1.2 million training images**, 1,000 categories (dogs, cats, cars, instruments, etc.).
- Images are high resolution and natural, collected from the internet.
- Became the **standard benchmark** for large-scale image classification.
- AlexNet(2012), VGG(2014), GoogLeNet(2014), ResNet(2015) and many other CNN architectures were all **designed, tested, and compared** on ImageNet-1K.

Resnet-50

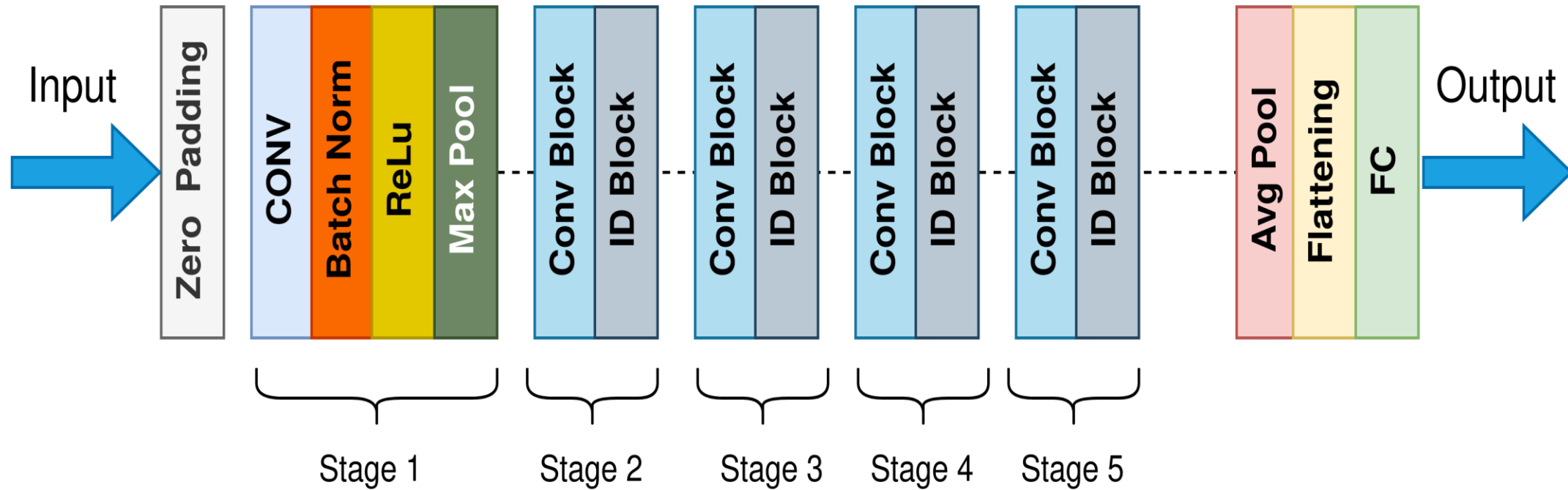
- **Input Image** → $224 \times 224 \times 3$ RGB image
- **Conv Layer + Max Pooling** → 7×7 convolution (stride 2) + 3×3 max pooling
- **Conv2_x** → 9 convolutional layers
- **Conv3_x** → 12 convolutional layers
- **Conv4_x** → 18 convolutional layers
- **Conv5_x** → 9 convolutional layers
- **Global Average Pooling** → reduces each feature map to a single value
- **Fully Connected Layer** → 1000 output units (for ImageNet classes)
- **Output** → Class probabilities (via Softmax)

Resnet-50



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ResNet50 Model Architecture



76% Top 1 accuracy, 93% Top 5 accuracy



Object Detection

- COCO Dataset (Common Objects in Context)
- Released by Microsoft in 2014.
- Contains **330,000 images**, with **80 object categories** and over **1.5 million labeled instances**.
- Categories include **people, animals (dog, cat, cow, etc.), vehicles, furniture, food, and more**.
- Each object is annotated with a **bounding box and class label** (unlike ImageNet, which is only classification).
- YOLOv5 can draw boxes around dogs, people, cars, etc.

YOLOv5 Architecture Overview:



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- **Input:** Image of arbitrary size (commonly 640×640 RGB)
- **Backbone:** Next slide
- **Neck/PAN (Path Aggregation Net)**
 - Merges features from shallow (early) and deep (later) layers.
 - Improves localization and detection of small objects.
- **Head/Detection Head:**
 - Several Conv layers (can be thought of like FC layers applied spatially) that predict:
 - **Bounding boxes**
 - **Objectness score**
 - **Class probabilities**
- **Output:** For each detected object: (x, y, w, h, confidence, class)



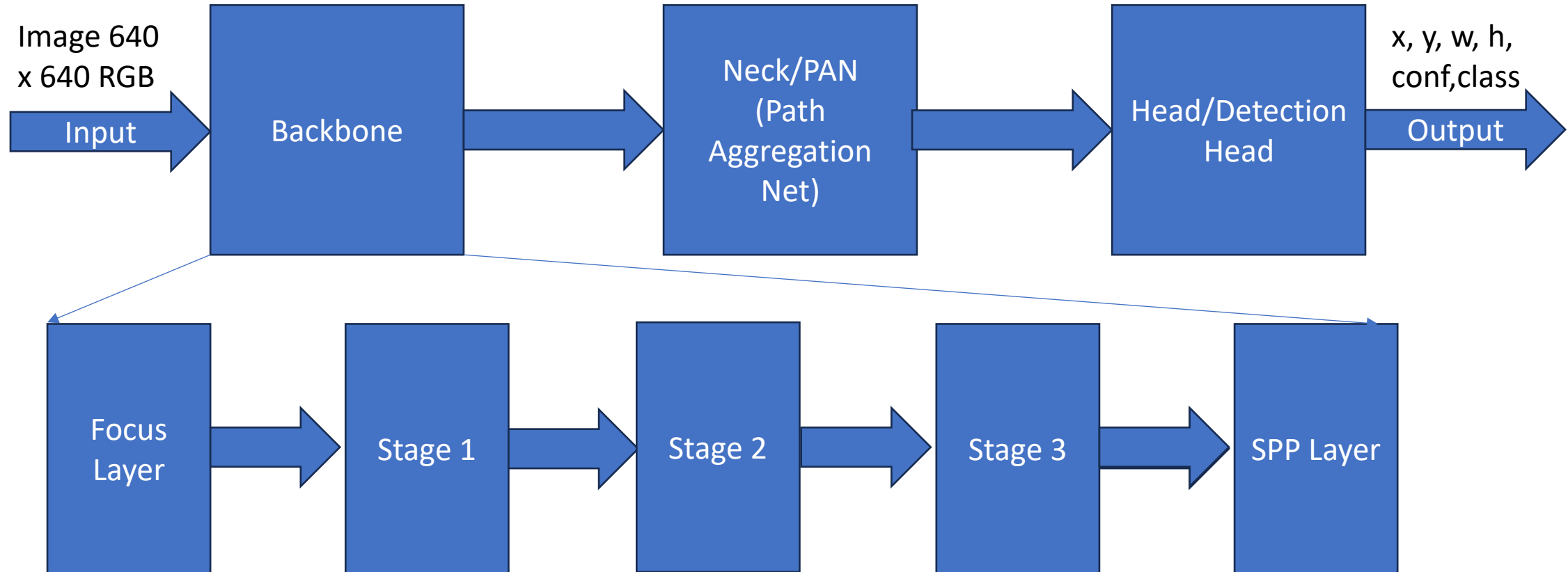
Backbone Architecture Overview

- **Focus Layer:**
 - Cleverly downsamples the input image (similar effect to pooling) using a convolution operation.
 - Reduces spatial dimensions while increasing channel depth.
- **Stage 1 (Backbone – Bottleneck/Conv):**
 - Conv Layer + 3 convolution layers (with skip connections)
 - Extracts low-level features like edges and textures.
- **Stage 2 (Backbone):**
 - Conv Layer + 6 convolution layers (with skip connections)
 - Extracts more abstract features.
- **Stage 3 (Backbone):**
 - Conv Layer + 9 convolution layers (with skip connections)
 - Captures higher-level, semantic features.
- **SPP Layer (Spatial Pyramid Pooling):**
 - Multi-scale pooling (5×5 , 9×9 , 13×13) to capture features at multiple receptive fields.
 - Helps detect objects of different sizes

Yolov5-S



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mAP@0.5: ~55–56%, mAP@0.5:0.95: ~36%, Precision: ~56%, Recall: ~37%



Model Comparison



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Model	Dataset	Backbone	Neck (PAN)	Head	Total Layers (approx.)
Perceptron: Linear Classifier	MNIST	N/A	N/A	Yes	1
LeNet-5: Simple Image Classification	MNIST	Yes	N/A	Yes	~7
ResNet-50: Modern Image Classification	ImageNet-1K	Yes	N/A	Yes	50
YOLOv5: Modern Image Detector	COCO	Yes	Yes	Yes	~70

